KONRAD ANAND

Edinburgh, UK **konradanand.github.io**

About me

I am a postdoc in the Department of Computer Science at the University of Oxford, hosted by Leslie Goldberg. I work in probability, combinatorics, and algorithms, particularly in relation to counting, sampling, and randomization.
I completed my Ph.D. in Mathematics at the School of Mathematical Sciences at Queen Mary University of London, supervised by Mark Jerrum. I completed my M.Sc. in Computer Science at McGill University, supervised by Luc Devroye.

Academic Positions

2026 – **Postdoc in Computer Science**, *University of Oxford*, Oxford, UK.
Hosted by Leslie Goldberg (cs.ox.ac.uk/people/leslieann.goldberg/)

July 2026 **Visiting Appointment in Mathematics**, *Kings College London*, London, UK.
Hosted by Alexandre Stauffer (sites.google.com/site/alexandrestauffer)

2024 – 2026 **Postdoc in Informatics**, *University of Edinburgh*, Edinburgh, UK.
Hosted by Heng Guo (homepages.inf.ed.ac.uk/hguo/)

Education

2020 – 2024 **PhD in Mathematics**, *Queen Mary University of London*, London, UK.
Supervised by Mark Jerrum (www.maths.qmul.ac.uk/∼mj/)
Thesis: *Perfect sampling via spatial mixing*

2018 – 2020 **M.Sc. in Computer Science**, *McGill University*, Montreal, Canada.
Supervised by Luc Devroye (luc.devroye.org)
Thesis: *Probabilistic Analysis of RRT Trees*, arXiv:2005.01242 (2020)

2012 – 2018 **B.Sc. Honours in Mathematics**, *McGill University*, Montreal, Canada.
Graduated with First-Class Honours

Papers

Submitted **Linear time approximation of the TV distance between product distributions**.
Konrad Anand, Alistair Benford, Heng Guo
– arXiv:2607.27088

Submitted **Simulating Gaussian boson sampling on graphs in polynomial time**.
Konrad Anand, Zongchen Chen, Mary Cryan, Graham Freifeld, Leslie Ann Goldberg, Heng Guo, Xinyuan Zhang
– arXiv:2511.16558

RANDOM 2025 **Sink-free orientations: a local sampler with applications**.
Konrad Anand, Graham Freifeld, Heng Guo, Chunyang Wang, Jiaheng Wang
– arXiv:2502.05877

SOCG 2025 **Rapid mixing of the flip chain over non-crossing spanning trees**.
Konrad Anand, Weiming Feng, Graham Freifeld, Heng Guo, Mark Jerrum, Jiaheng Wang
– arXiv:2409.07892

AIHPD 2024 **Perfect Sampling of q-Spin Systems on $\mathbb{Z}^2$ via Weak Spatial Mixing**.
Konrad Anand and Mark Jerrum
– arXiv:2302.07821

ICALP 2024 **Approximate Counting for Spin Systems in Sub-Quadratic Time**.
Konrad Anand, Weiming Feng, Graham Freifeld, Heng Guo, and Jiaheng Wang
– arXiv:2306.14867

RANDOM 2023 **Perfect Sampling for Hard Spheres from Strong Spatial Mixing**.
Konrad Anand, Andreas Göbel, Marcus Pappik, and Will Perkins
– arXiv:2305.02450

SICOMP 2022 **Perfect Sampling in Infinite Spin Systems via Strong Spatial Mixing**.
Konrad Anand and Mark Jerrum
– arXiv:2106.15992

Talks

Mar. 2026 **Perfect sampling for hard spheres from strong spatial mixing**, *Point Processes and their Applications*.
Lille

Mar. 2026 **Sink-free orientations: a local sampler with applications**, *Workshop on the Combinatorial, Algorithmic and Probabilistic aspects of Partition Functions*.
CWI Amsterdam

Mar. 2025 **Sink-free orientations: a local sampler with applications**, *Warwick Applied Probability Seminar*.
University of Warwick

Sep. 2024 **The distribution of long-range correlations under weak spatial mixing**, *Mixing Times Workshop with Applications to Theoretical Computer Science*.
University of Bath

May. 2024 **Lazy depth-first sampling on infinite spin systems**, *Warwick Applied Probability Seminar*.
University of Warwick

Mar. 2023 **Lazy depth-first sampling on infinite spin systems**, *PGR Meeting*.
Queen Mary University of London

Dec. 2022 **Lazy depth-first sampling on infinite spin systems**, *Counting and Sampling: Algorithms and Complexity*.
Dagstuhl Seminar 22482

Nov. 2022 **Lazy depth-first sampling on spin systems**, *QMUL Combinatorics Seminar*.

Sep. 2022 **Lazy depth-first sampling**, *Processes on Random Geometric Graphs*.
University of Cologne

Meetings

Mar. 2025 **Point Processes and their Applications**, *Lille*.
Université de Lille

Dec. 2025 **FOCS**, *Sydney*.
Australia

Aug. 2025 **XXVIII Brazilian School of Probability**, *Belo Horizonte*.
Universidade Federal de Minas Gerais

Jun. 2025 **Symposium on Computational Geometry**, *Kanazawa*.
CG Week

May. 2025 **Conference on Mixing Times between Probability, Computer Science and Statistical Physics**, *Trieste*.
International Centre for Theoretical Physics

Mar. 2025 **Dynamics of random systems—UK Easter Probability Meeting**, *Durham*.
Durham University

Mar. 2025 **Workshop on the Combinatorial, Algorithmic and Probabilistic aspects of Partition Functions**, *Amsterdam*.
CWI Amsterdam

Sep. 2024 **Mixing Times Workshop with Applications to Theoretical Computer Science**, *Bath*.
University of Bath

Dec. 2022 **Counting and Sampling: Algorithms and Complexity**, *Dagstuhl*.
Dagstuhl Seminar 22482

Sep. 2022 **Processes on Random Geometric Graphs**, *University of Cologne*.
Summer school focused on new developments on dynamics on spatial random networks

July 2019 **Nice Summer School**, *Université de Nice Sophia Antipolis*.
Summer school focused on random walks, Markov chains, and random graphs

July 2019 **École d'été Graphes et Arbres Aléatoires**, *Centre International de Rencontres Mathématiques* .
Summer school focused on planar maps, continuum trees and random graphs

Awards

2023 **PGR Meeting Best Talk – 3rd Prize**, *Queen Mary University of London*.

2020 – 2024 **PGR Studentship**, *Queen Mary University of London*.

2017 **NSERC USRA**, *McGill University*.
Undergraduate research scholarship in mathematics

2017 **Supplément aux bourses de 1er cycle du CRSNG**, *McGill University*.
Undergraduate research scholarship in mathematics

Teaching

2022 **Teaching Assistant**, *King's College London*.
7CCMFM01: Probability Theory

2021-2022 **Teaching Assistant**, *Queen Mary University of London*.
MTH4107 / MTH4207: Introduction to Probability
MTH5105: Differential and Integral Analysis
MTH5114: Linear Programming and Games
MTH6105: Algorithmic Graph Theory

2016-2020 **Teaching Assistant**, *McGill University*.
MATH 235: Algebra 1
MATH 589: Advanced Probability 2
COMP 251: Algorithms and Data Structures
COMP 252: Honours Algorithms and Data Structures
COMP 360: Algorithm Design
Undergraduate Mathematics Help Desk